PFAS binding to serum albumin: a standalone primer and review

What this is. A self-contained introduction to PFAS–albumin binding, written for someone starting from zero, followed by a review of what the literature actually establishes and what our own analysis added. It assumes no prior knowledge of binding chemistry.

Why it exists separately. The albumin material in `REPORT.md` (§6.1–6.3) is written as evidence inside a larger argument about species differences. This document builds the subject from the ground up instead.

Every code block runs as-is on plain Python 3 — no imports from this repository, no packages to install. Copy and run them.

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Part I — Foundations

1. Why anyone cares

The chain of reasoning that makes albumin binding a toxicokinetics question, in four steps:

1. PFAS circulate in blood. Most of what is in blood is **not floating free** — it is stuck to proteins.
2. Only the **free** (unbound) fraction can be filtered by the kidney, cross a membrane, or reach a receptor. Bound PFAS is, for that moment, inert and un-excretable.
3. So the unbound fraction, written **f_u**, sets how fast the body can clear the compound. A smaller f_u means slower clearance and a longer half-life.
4. Albumin is the most abundant protein in plasma and the one most PFAS stick to. So "how tightly does PFAS X bind albumin?" is upstream of "why does PFOA have a 3.8-year half-life in humans?"

That is the motivation. Whether the chain actually holds up quantitatively is §13, and the answer is "partly".

2. What albumin is

Human serum albumin (HSA) is a single protein, 66,463 g/mol, about 585 amino acids, shaped like a heart in three domains. Facts worth carrying:

- **It is abundant.** Plasma holds 35–50 g/L of it. At 66,463 g/mol that is **526–752 µmol/L**; 600 µM is the usual working figure. Nothing else in plasma comes close.
- **Its day job is carrying fatty acids.** Fatty acids are oily and do not dissolve in blood; albumin ferries them. It has **seven** recognised fatty-acid sites, plus two classic drug sites (Sudlow I and II), plus a single tryptophan (Trp214) whose fluorescence makes a convenient probe.
- **PFAS look like fatty acids.** A perfluorinated chain with an acid head is structurally a fatty acid with fluorines where hydrogens would be. It binds in the same pockets. That is the whole mechanism in one sentence.
- **Different species have their own version** — bovine (BSA), rat (RSA), porcine (PSA). They are similar but not identical, which matters when a study uses BSA as a cheap stand-in for HSA (§10).

The key structural consequence: **albumin does not have "a" binding site.** It has several, of different affinity. Most reported numbers lump them into one effective constant, and a lot of the apparent disagreement in the literature is really disagreement about which site got measured (§8).

3. The vocabulary, built from one reaction

Everything starts with one reversible reaction — ligand L (the PFAS) meets protein P (albumin) and sticks:

```
L + P  <=>  LP
```

At equilibrium the rates balance, and the ratio of concentrations is fixed:

K_d — the dissociation constant

```
K_d = [L][P] / [LP]          units: concentration, usually µM or M
```

K_d is the **ligand concentration at which half the sites are occupied**. That is the intuition to keep. It has units of concentration because it *is* a concentration.

- **Small K_d = tight binding** (it takes very little ligand to fill the sites)
- **Large K_d = weak binding**

K_a — the association constant is simply its reciprocal:

```
K_a = 1 / K_d                units: M^-1  (per molar)
```

Large K_a = tight binding. Papers use one or the other more or less arbitrarily, which is the first thing that makes the literature hard to read. A paper reporting K_a = 10^6 M^{-1} and one reporting K_d = 1 µM are saying the same thing.

```
# Converting between them. Run this.
def ka_from_kd(kd_uM):
    """Kd in micromolar -> Ka in M^-1."""
    return 1.0 / (kd_uM * 1e-6)

def kd_from_ka(ka_M):
    """Ka in M^-1 -> Kd in micromolar."""
    return 1.0 / ka_M * 1e6

print(f"Kd =  1 uM   -> Ka = {ka_from_kd(1.0):.3g} M^-1")
print(f"Kd = 100 uM  -> Ka = {ka_from_kd(100.0):.3g} M^-1")
print(f"Ka = 2.7e5   -> Kd = {kd_from_ka(2.7e5):.3g} uM")
```

n — the number of binding sites, sites per protein molecule. Usually fitted alongside K_d, often badly. A value below 1 means the fit found less than one site's worth of capacity, which tells you it is an *effective lumped parameter*, not a count of pockets.

f_u — the unbound fraction, the number TK actually wants: what share of the total PFAS in plasma is free.

4. Three views of the same number

This is the part worth slowing down on, because the same physical fact gets written three ways and people talk past each other.

View 1 — occupancy (the chemist's view)

What fraction of the *protein's sites* are filled?

```
occupancy = [L_free] / (K_d + [L_free])
```

At physiological PFAS levels this number is tiny — albumin is almost entirely empty.

View 2 — the unbound fraction (the toxicokineticist's view)

What fraction of the *ligand* is free? Starting from K_d = [L][P]/[LP] and substituting, the free-ligand term cancels and you get something clean:

```
f_bound = [P] / ([P] + K_d)        and     f_u = 1 - f_bound
```

where [P] is the concentration of *binding sites* = n × [albumin].

The cancellation only works when there is far more protein than ligand ([L] << [P]). In plasma that holds easily. In most laboratory assays it does not, and that is §6.

```
# View 2. How tight must binding be to explain what we see in blood?
ALB_uM = 600.0          # albumin in human plasma, 40 g/L / 66463 g/mol

def f_bound(kd_uM, n=1, alb_uM=ALB_uM):
    return (n * alb_uM) / (n * alb_uM + kd_uM)

def kd_needed(target_bound, n=1, alb_uM=ALB_uM):
    """Kd that would produce a given bound fraction."""
    return n * alb_uM * (1.0 / target_bound - 1.0)

print("Kd required to reach a given bound fraction:")
for f in (0.95, 0.99, 0.999):
    print(f"  {100*f:5.1f}% bound ->  n=1: {kd_needed(f,1):7.1f} uM"
          f"   n=6: {kd_needed(f,6):7.1f} uM")

print("\nMeasured PFOA (Chen 2025, Kd = 2.57 uM):")
for n in (1, 3, 6):
    print(f"  n={n}: {100*f_bound(2.57, n):.3f}% bound")
```

PFOA is measured as >99% bound in human plasma. Run the block: that requires K_d below about 6 µM at one site. **Any reported constant much weaker than that cannot reproduce what is seen in blood** — which is a free, powerful filter on a messy literature.

View 3 — the partition coefficient (the modeller's view)

A mass-balance or PBPK model does not want an affinity, it wants a **capacity**: how many litres of water's worth of PFAS one kilogram of albumin can hold. That is a partition coefficient, written **D** (or K_protein/water):

```
D = (amount per kg protein) / (amount per L water)
```

Views 1–3 are the same measurement in different clothes. §12 gives the exact bridge between them.

> ***Pause and check.*** *Before reading on: if K_d for compound A is 1 µM and for compound B is 100 µM, which binds more tightly, and by how much? And which would you expect to be cleared from blood faster, all else equal?* *A binds 100× more tightly. B should clear faster, because more of it is free at any moment. "All else equal" is doing a lot of work — §13.*

Part II — Why the literature disagrees

5. How people actually measure this

Six methods dominate. They do not measure quite the same thing, and knowing which was used tells you most of what you need about a number's reliability.

method	what it does	what it really gives	watch out for
Equilibrium dialysis	separates free from bound across a membrane	the reference standard; a true K_d	slow; PFAS stick to the apparatus
Solid-phase microextraction (SPME)	a coated fibre samples only free ligand	free concentration over a wide range	needs careful calibration

method	what it does	what it really gives	watch out for
Fluorescence quenching	watches Trp214's glow dim as ligand binds	binding *near that one tryptophan*	reports only the site it can see
Probe displacement	measures how much ligand it takes to evict a dye from a known pocket	site-resolved constants	depends on the probe's own affinity
Differential scanning fluorimetry (DSF)	measures how much the ligand stabilises the protein against melting	an apparent affinity **at the melting temperature**, ~60–70 °C	not a 37 °C equilibrium constant
Ultrafiltration / nanoESI-MS	physically separates, or counts complexes	varies	non-specific losses

6. The spread, and what causes it

For a single PFAS–protein pair the reported association constants span **four to five orders of magnitude**:

chemical	protein	n studies	K_a range (M^{-1})	spread
PFBS	HSA	5	$116 – 6.5 \times 10^6$	55,800×
PFBA	HSA	4	$20.4 – 1.1 \times 10^6$	54,000×
PFOS	HSA	6	$395 – 7.6 \times 10^6$	19,300×
PFOA	HSA	9	$103 – 1.98 \times 10^6$	19,200×

The natural guess is "different methods". **That is not the main cause** — equilibrium dialysis alone spans 3.2×10^2 to ~10^6, and fluorescence quenching spans 10^2 to 2×10^6 by itself. The real drivers are four things, and all four are knowable from a paper's methods section.

7. The four axes that move a binding constant

Axis 1 — the ligand:protein ratio (the big one)

Write ***nu*** for the PFAS:protein molar ratio.

situation	[PFAS]	*nu*
human background, 4 ng/mL	0.0097 µM	1.6×10^{-5}
high-exposure community	0.48 µM	8×10^{-4}
fluorochemical worker	12 µM	0.02
typical DSF assay	1000 µM	**1.7**
typical fluorescence assay	80 µM vs 2 µM HSA	**40**

Laboratory assays run **three to six orders of magnitude** above the body.

Why it matters: at low *nu*, ligand finds the few tight ("specific") sites. At high *nu* those fill up and the rest smears onto weak, non-saturable ("non-specific") surface. The same protein and the same compound give **two different constants** depending on which regime you are in.

This used to be an inference. **Qin et al. 2024 measured both, separately:**

compound	specific log D	non-specific log D	ratio
PFBA	2.44	1.97	3.0×
PFHxA	3.26	2.69	3.7×
PFHpA	4.12	3.47	4.5×
PFOA	4.58	3.88	5.0×

Specific binding dominates below *nu* ~ 0.01 (the body); non-specific takes over above *nu* ~ 1 (the assay). **A constant carried from assay to body without asking which regime produced it is wrong by 3–5×**, and that is the gap within one method in one laboratory — a floor, not a ceiling.

```
# Axis 1, made concrete. Saturation drives the apparent constant down.
def apparent_D(D_trace, Cw_uM, kd_uM):
    """A partition coefficient measured at free concentration Cw."""
    return D_trace / (1.0 + Cw_uM / kd_uM)

D0, kd = 1e5, 2.57          # trace-limit D, and PFOA's Kd in uM
print("free PFAS (uM)   apparent D   fraction of the true value")
for cw in (0.01, 0.1, 1, 10, 100, 1000):
    d = apparent_D(D0, cw, kd)
    print(f"{cw:>12}   {d:10.0f}   {d/D0:.3f}")
```

Axis 2 — temperature

DSF reads affinity off a **melting transition**, around 60–70 °C. Its own authors say so. DSF constants rank congeners usefully and should not be carried into a kinetic model as 37 °C affinities.

Axis 3 — ionic strength

Gao et al. 2025 is the only chain-length series measured across added salt. Affinity falls **monotonically** with ionic strength, by roughly **1.5–2×** between a 20 mM buffer and the ~150 mM of plasma, and the shift is independent of chain length and head group. A constant measured in dilute buffer overstates binding before any other correction.

Axis 4 — which site

See §8. This one is routinely invisible because most papers report one number without saying which pocket produced it.

8. The site problem

Chen & Guo 2009 measured five PFAS at **three** sites with three probes. The full grid, in K_a (M^{-1}):

compound	Trp214	Sudlow I	Sudlow II
PFBA	no binding	1.1×10^6	no binding
PFBS	no binding	2.2×10^6	6.5×10^6
PFOA	**2.7×10^5**	binds, not quantified	binds, not quantified
PFOS	**2.2×10^4**	binds, not quantified	7.6×10^6
PFDoA	binds, not quantified	no binding	1.2×10^6

Read across a row: **PFOS binds Sudlow II 345× more tightly than Trp214.** Same paper, same protein, same compound.

The bold cells are the ones regulatory summaries carry — and the authors themselves identify the tryptophan site as **the weakest of the three**. Quoting it makes PFOS look like a weak binder (K_d 45 µM) when its best site gives 0.13 µM.

A K_d quoted without its site is as incomplete as one quoted without its n.

The same paper also supplies the control the field usually lacks. For the same three probes the authors measured plain, non-fluorinated fatty acids: octanoic acid binds Sudlow II at 1.1×10^7 M^{-1} — **tighter than every PFAS in the grid**. PFAS occupy the fatty-acid pockets without outcompeting the fatty acids, which is why the "PFAS displace endogenous ligands" story needs care.

> ***Pause and check.*** *A paper reports "K_a = 2.2×10^4 M^{-1} for PFOS–HSA by fluorescence quenching". Name three things you would want to know before using it, and say roughly what each could do to the number.* *Which site (up to 345×), the ligand:protein ratio (3–5×, or more), the buffer ionic strength (1.5–2×). Temperature too, if the method were DSF.*

Part III — What is actually known

9. Structure: chain length and head group

Chain length is not monotonic. This is the finding that most surprises people. Every dataset that reaches far enough eventually turns over. *Where* it turns over is contested, and §9a settles that; start here with the shape.

Chen et al. 2025, equilibrium dialysis, HSA at 1 µM — the cleanest series available, K_d in µM (**lower = tighter**):

perfluorinated carbons	PFCA	K_d (µM)	PFSA	K_d (µM)
3	PFBA	84.9		
4	PFPeA	16.3	PFBS	15.6
5	PFHxA	14.0	PFPeS	17.2
6	PFHpA	6.2	PFHxS	**1.31**
7	**PFOA**	**2.57**	PFHpS	1.68
8	PFNA	3.75	**PFOS**	**0.94**
9	PFDA	3.64	PFNS	6.56
10	PFUnDA	4.73	PFDS	1.88
11	PFDoDA	15.8		
12	PFTrDA	31.4	PFDoS	1.29
13	PFTeDA	13.1		

Binding tightens from PFBA to about PFOA, then **weakens again**. The mechanism is not in dispute: albumin's fatty-acid sites are channels of finite depth, and past some length the chain no longer fits and has to bend or leave part of itself in water. A bilayer has no such limit, which is why membrane partitioning never turns over (§9a).

Two consequences:

- **A structure–activity model linear in carbon number is wrong at both ends.**
- The compounds that bind albumin most tightly are not the longest ones. Long-chain persistence is not an albumin story.

What Chen's table gets wrong is the *position* of the turn. Taken at face value it puts the maximum at seven perfluorinated carbons, and §9a shows that is about three carbons too early.

The head group is worth about a log unit. At matched chain length the sulfonate binds harder than the carboxylate — PFHxS 1.31 vs PFHpA 6.2 µM at six fluorinated carbons; PFOS 0.94 vs PFNA 3.75 at eight.

Albumin stops being the main carrier at long chains. Fischer et al. 2024 measured *gamma*-globulin alongside albumin: globulin binds 2–3 log units weaker at short chains, catches up steadily, and for PFTeDA binds *more* tightly than albumin. Treating "protein binding" as "albumin binding" degrades exactly where the persistent compounds live.

9a. Where the optimum actually is

Four series in `db/protein_binding.csv` bracket a maximum — they have measured points on *both* sides of their highest value — and they do not agree:

dataset	method	nCF range	peak	span (nCF 4–9)	reversals
Fischer 2024 HSA	SPME depletion	4–13	**10**	2.57	1
Fischer 2024 BSA	SPME depletion	5–13	**10**	1.98	1
Fischer 2024 *gamma*-globulin	SPME depletion	5–13	**11**	1.94	1
Fischer 2024 whole serum	SPME depletion	4–13	**10**	1.41	2
Allendorf 2019 BSA	equilibrium dialysis, 37 °C	3–10	**9**	1.30	1
Bischel 2011 BSA	equilibrium dialysis	4–11	6	0.83	1
Chen 2025 HSA	equilibrium dialysis	3–15	7	0.80	4
Starnes 2025 HSA	DSF	3–10	6	0.67	3
Muensterman 2026 HSA	biomimetic chromatography	3–17	7	0.60	1

Three further series — Gao 2025, Qin 2024's nonspecific term, Starnes 2024 — never reach a maximum at all, which is not a neutral result: Gao is still climbing at nCF 9 and Qin at nCF 10, and both would have to be *falling* there if the optimum were at 6–8.

Sort that table by span and the peak sorts with it. Spearman *rho* between dynamic range over the shared window and peak position is **+0.80** across the nine bracketed series, from eight laboratories and six methods. Every series spanning more than 1.3 log units peaks at nCF 9–11; every series spanning less than 0.85 peaks at 6–7. A curve that cannot resolve three orders of magnitude cannot locate a maximum inside them.

Muensterman 2026 is the newest and the largest — 83 compounds, the only dataset here reaching nCF 17 — and it lands exactly where the relationship predicts: the narrowest range in the table, 0.60 log units, and a peak at nCF 7. It is also the one measurement in this section that is not an equilibrium method at all. Retention on an albumin-coated column is calibrated against pharmaceuticals of known partition coefficient, so a compressed response is what the technique produces rather than something that happened to it.

The compression from §11b is the wrong tool for this. An argmax survives any strictly increasing transform, so rescaling Fischer's curve by 0.82, 0.5 or 0.25 leaves its peak exactly where it was. Chen and Fischer disagree in shape, not in scale. Regressing Chen on Fischer makes that concrete: the slope is 0.15 (the same compression, seen from the other side) but r^2 is only 0.14, and the residuals are positive across nCF 6–10 and negative at 11–12.

The test that decides it. Fischer measured four sorbents in one experiment with one method — HSA, BSA, *gamma*-globulin and whole serum. If the fall after nCF 10 were the fiber or the chemistry giving out, it would give out at a fixed *level*, because that is what a ceiling is:

sorbent	peak nCF	log D at the peak	fall to nCF 13
HSA	10	5.18	1.31
BSA	10	5.17	1.30
gamma-globulin	**11**	4.81	0.81
whole serum	10	3.87	0.87

The peak *levels* span 1.31 log units. The peak *positions* span one carbon. And globulin — a different protein in the same tube on the same fiber — peaks one CF_2 later than albumin, which is what two cavities of slightly different depth look like and is not what a shared artefact looks like. The two albumins agree to 0.01 log units at the peak and to 0.00 at nCF 13, the far end of a 20-fold fall.

Chen's long-chain tail fails on its own terms. K_D goes 4.73 -> 15.8 -> 31.4 -> 13.1 -> 10.5 µM across nCF 10, 11, 12, 13, 15: weaker, weaker, then *tighter* again. With Chen's own standard errors that reversal is real (31.4 ± 5.1 against 13.1 ± 3.0), and a binding cavity cannot do it. Meanwhile the membrane–water coefficient Chen measured on the same compounds in the same study rises smoothly all the way to nCF 15 (+0.19 to +0.47 per CF_2, no reversal). The long-chain compounds were present and quantifiable; what fails is the dialysis.

The ligand:protein ratio is *not* what separates them, and I had that wrong. **It was tempting: Fischer works at <= 0.004, Chen at up to 2, DSF at micromolar to millimolar. But Bischel 2011 ran equilibrium dialysis against 200 µM BSA with 0.009–0.68 µM PFAA — a ratio of at most 0.0034**, *lower than Fischer's* — and still produced the second most compressed series in the table and a peak at nCF 6. Allendorf held its ratio below 0.1 and landed at nCF 9. So the regime matters for what a constant *means* (§7 Axis 1) but does not predict where a series puts its maximum. Dynamic range does.

The solubility caveat still stands on its own: the 2026 Starnes review says of its own method family that "the poor aqueous solubility of many PFAS congeners, particularly long-chain PFAAs (C12 and above), prevents accurate measurement of their binding in the assay." C12 is nCF 11 — exactly where the datasets part company.

The answer. Averaging Fischer's two albumins and reading the top of the curve rather than a single point:

nCF	compound	HSA	BSA	mean	below the peak
8	PFNA	4.49	4.54	4.52	-0.66
9	PFDA	4.73	4.81	4.77	-0.41
10	**PFUnDA**	5.18	5.17	**5.17**	**0.00**

nCF	compound	HSA	BSA	mean	below the peak
11	PFDoDA	5.05	5.12	5.08	-0.09
12	PFTrDA	4.54	4.47	4.50	-0.67
13	PFTeDA	3.87	3.87	3.87	-1.30

> ***The optimum is a plateau at nCF 10–11 — PFUnDA and PFDoDA, the C11 and C12 acids — centred on nCF 10 at log D ~ 5.2 L_water per kg protein. Widen the tolerance to half a log unit and nCF 9 (PFDA) joins; nothing else does.***

Confidence, separated out, because the three parts are not equally strong:

- *That an optimum exists and is not an artefact* — **strong**. Two proteins, 0.01 log apart at the peak; position invariant across three sorbents 1.3 log units apart in level.
- *That it sits at 10 rather than 9 or 11* — **moderate**. The top is flat and nobody publishes per-compound error bars on this stretch.
- *That it is **not** at nCF 6–8* — **strong**, and this is the part that matters. Fischer's albumins peak at 10 and its *gamma*-globulin at 11; Allendorf's equilibrium dialysis at 37 °C peaks at 9 with a smooth curve; Gao's fluorescence quenching is still climbing at 9; Qin's nonspecific SPME term is still climbing at 10. Four methods, four labs, and a maximum at 7 needs all four wrong in the same direction. Even Chen's own quadratic vertex, fitted through its peak and neighbours, lands at 8.2 rather than at its argmax of 7.

The sulfonate optimum is not located. Fischer's PFSA series ends at nCF 10 (PFDS) still at or near its top, so all that can be said is nCF >= 9. No dataset here carries a C12 sulfonate.

Why this is invisible in vivo, and why that is not a contradiction

Three statements that sound incompatible are all true:

- tissue binding has no chain-length optimum — Ryu 2024's unbound fractions, five tissues × four species, fall monotonically to nCF 13 at -0.50 per CF_2 with no reversal;
- membrane partitioning has no chain-length optimum — Chen's own K_MW, to nCF 15;
- **albumin binding does** — Fischer, at nCF 10.

They are consistent because past the optimum the long-chain PFCAs do not stop binding. They stop binding *to albumin* and go on partitioning into lipid. Anything measured on whole tissue, or on whole serum, sums the two and sees the sum keep rising. So an optimum found on purified protein is not refuted by an in vivo series that shows none — and the reverse inference, "half-lives keep rising with chain length, so albumin affinity must too", does not follow either.

What it changes

1. **Every regulated PFCA is on the rising limb.** PFOA sits 0.70 log units — a factor of 5 — below the albumin optimum; PFNA 0.69; PFDA 0.45. That is the regime where affinity is most sensitive to chain length, which is where a QSAR has the most to say and the most to get wrong.
2. **A QSAR fitted on nCF <= 9 extrapolates straight through the optimum.** Fischer's HSA limb runs at +0.49 per CF_2 over nCF 4–9. Extending that line over-predicts nCF 11 by 3×, nCF 12 by 34× and nCF 13 by 487×. Almost no training set with toxicity values reaches nCF 12.
3. **It does not rescue albumin in the mass balance** (§13). The compounds that failed — PFHxA at nCF 5, PFOA at 7 — are on the rising limb, five and three carbons from the optimum. PFHxA needed log D 3.8–4.0 against 3.20 measured, and the optimum cannot be borrowed across five CF_2 groups.

Run it: `python3 scripts/albumin_chain_length_optimum.py` (`--figure` writes `figures/fig17_albumin_chain_length_optimum.png`).

10. Species: albumin is conserved, and that is the problem

Starnes et al. 2024 measured four albumins by one method, K_d in mM (lower = tighter):

compound	human	rat	bovine	porcine	rat/human
PFOA	0.79	0.40	0.99	0.37	**0.51×**

compound	human	rat	bovine	porcine	rat/human
PFOS	0.69	0.42	0.86	0.38	0.61×
PFBA	2.64	1.90	3.06	1.28	0.72×

The differences are statistically significant and **biologically trivial**: highest-to-lowest never exceeds ~4×, usually under 2×.

Now the punchline. **Rat albumin binds PFOA about twice as tightly as human albumin — and the rat clears PFOA roughly a thousand times faster.** The direction is wrong and the magnitude is off by two and a half orders of magnitude. Albumin affinity cannot be the cause of the species difference in elimination.

Three independent matrices now agree:

- **Albumin** (Starnes 2024, Fischer 2024): <=4× across four species.
- **Whole tissue** (Ryu 2024, 16 PFAS × 5 tissues × 4 organisms): log f_unbound correlates across every species and tissue pair at R^2 >= 0.97.
- **Whole plasma** (Qin 2024): the human-versus-trout difference is explained by **composition** — human plasma is 42.25 mL/L protein and 4.46 mL/L lipid, trout 15.46 and 7.08 — not by protein identity. BSA is a good surrogate for human plasma.

Species differ in what their plasma and tissues are made of, not in how their proteins bind.

> ***A gap worth knowing about.*** *There is* ***no in vitro measurement of PFAS binding to mouse serum albumin*** *— not in our 865-row database, not in PubMed. The mouse is the animal most of the toxicology runs on.*

11. The best current numbers

If you need one dataset, use **Chen et al. 2025 Table S5**: 60 PFAS, equilibrium dialysis, HSA fixed at 1.0 µM with ligand at 0.0625–2 µM, so *nu* = 0.06–2. That is one to two orders of magnitude closer to physiological than the DSF and fluorescence datasets, and it is the widest coverage at that ratio.

The sanity check that matters: **56 of its 60 compounds reproduce >99% plasma binding** at six sites. The whole dataset lands in the physiologically possible band, which is not true of the DSF or 1954 surfactant literature.

For comparison, scoring the older literature against the same requirement:

source	K_d (µM)	predicted bound (n=6)	
Chen 2025 (PFOS)	0.94	100%	yes
Chen 2025 (PFOA)	2.57	99.9%	yes
Chen & Guo 2009 (PFOS, Sudlow II)	0.13	100%	yes
Bischel 2010	0.7–5	~100%	yes
Qin 2010	21	99.4%	yes
Beesoon & Martin 2015 (n-PFOA)	100	97.3%	borderline
Han 2003	400	90.0%	no
Starnes 2024 (DSF)	790	82.0%	no
Klevens & Ellenbogen 1954	3100	53.7%	no

11a. The two best datasets appear to disagree. They do not.

Chen 2025 and Fischer 2024 are the two cleanest albumin datasets here, both at defensible ligand:protein ratios. Converting Chen's PFOA (K_d = 2.57 µM) at one site gives log D = 3.77. Fischer measured **4.48**. About 5× apart — which looked, at first, like a real problem.

It is not a disagreement. **They measure the two different binding regimes**, and each lands on the matching term of the one study that measured both.

Putting all four on the same basis (mass, L_water/kg_protein; Qin shifted -0.135 from his volume basis):

compound	Fischer	Qin specific	gap	Chen	Qin non-specific	gap
PFHxA	3.20	3.13	+0.07	3.03	2.56	+0.48
PFHpA	4.10	3.99	+0.11	3.39	3.34	**+0.05**
PFOA	4.48	4.45	**+0.03**	3.77	3.75	**+0.02**
PFHxS	4.77	4.89	-0.12	4.06	3.45	+0.62
PFOS	4.62	5.14	-0.52	4.20	4.04	+0.17

For PFOA the two matches are **0.03 and 0.02 log units** — 7% and 5%. Three laboratories, three methods, two regimes, and each pair lands on the other.

Why it is not a stoichiometry artefact. The competing explanation was that Chen's conversion assumed n = 1 when it is really ~5, which would close the gap arithmetically (§12: log D = log K_a + log n - 1.823). But Chen *currently* sits on Qin's non-specific term to 0.02 log units. Raising his n to 5 would lift him onto Fischer and in doing so **break an agreement that presently holds to 5%**. You cannot have both. And n ~ 1 is what §12's backwards conversion found independently (n_implied = 1.01 for PFOA, 1.12 for PFHpA).

So which do you use? It depends entirely on the question:

- **A PBTK or plasma-binding calculation -> the specific constant** (Fischer, or Qin's specific term). The body runs at $nu < 0.01$.
- **An in vitro mass-balance model of a dosed assay -> the non-specific constant** (Chen, or Qin's non-specific term). The assay runs at $nu > 1$.

Getting it backwards costs about 5× for carboxylates and more for sulfonates.

One caveat that remains. The sulfonates reconcile less cleanly — Qin's own specific-to-non-specific separation is much larger for them (PFHxS 27×, PFOS 13×) than the Fischer–Chen gap, so something else is also moving there.

Reproduce: `python3 scripts/chen_fischer_reconciliation.py`

11b. Chen's chain-length profile is compressed, and that is the PFNA story

§11a flagged one residual: Chen's K_d puts PFNA (3.75 µM) *weaker* than PFOA (2.57 µM), where Fischer and Qin both put it at or above. It turns out not to be about PFNA at all.

First, it is not significant. Chen reports standard errors, so the question can be asked directly. On log D with the error propagated:

pair	difference	SE	p
PFOA vs PFHpA	0.382	0.110	**0.000**
PFOA vs PFNA	0.164	0.117	0.161
PFOA vs PFDA	0.151	0.127	0.233
PFOA vs PFUnDA	0.265	0.125	0.034

PFOA, PFNA and PFDA are **statistically indistinguishable**. Calling it an inversion overstates what the data support — PFOA is nominally tightest, and PFNA/PFDA sit in a plateau the measurement cannot separate.

Second, the real difference is dynamic range. Over the same compounds Chen's PFCA series spans **1.52 log units** and Fischer's spans **3.02**. PFNA is one symptom of that compression, not a separate fault.

A caution on what compression does and does not explain, because I got this wrong first time: Chen's apparent optimum sits at PFOA (nCF 7) and Fischer's at PFUnDA (nCF 10), and **compression is not the reason**. An argmax survives any strictly increasing transform, so shrinking a curve toward its mean leaves its peak exactly where it was. The two disagree in *shape*, and §9a works out which shape is right.

Third, what compresses it. Three candidate predictors of the shortfall (Fischer - Chen), each regressed across the series:

predictor	r^2, all 16	r^2, PFCA only		
affinity (Fischer log D)	**0.76**	**0.90**		
chain length (nCF)	0.41	0.56		
identifiability, \	log(K_d/[P])\		0.03	0.07

Affinity wins decisively. (The identifiability idea — that a single-site K_d is only well determined near the protein concentration — was my first guess and explains nothing; it is reported because testing it was the point.) Chain length is a real but weaker correlate, which is expected because affinity rises with chain length over most of the series; the two cannot be fully separated here.

The fitted slope is **0.82**: roughly four fifths of any true affinity difference is lost. A compression has a pivot, and this one sits at Fischer log D ~ 3.14 — below it, Chen should read *tighter*, with the gap changing sign. Leave-one-out prediction (each compound predicted from a fit excluding it) gives a mean miss of 0.23 log units, and for PFPeA, the one compound weak enough to test the sign flip, **-0.80 predicted against -0.81 observed**.

So it is a genuine compression about a pivot — not a uniform offset, not a few bad cells. Chen reads weak binders too tight and tight binders too weak, by a factor one line predicts across three log units.

What this means for using Chen:

- The **ranking** is sound at the short end. Use it up to about nCF 8.
- **Do not read a chain-length optimum off Chen alone.** Fischer puts it at nCF 10 and Chen at nCF 7. That difference is *not* the compression (see the caution above) — it is a separate failure in Chen's long-chain tail, where K_D reverses direction between nCF 12 and 13. §9a adjudicates it and comes down on nCF 10–11.
- The **PFNA cell is within noise of PFOA** and should not be treated as a finding in either direction.

The mechanism behind the affinity dependence is most likely that a tight binder's specific site is already substantially occupied at Chen's lowest ligand concentration, so the single-site fit is dragged toward the non-specific value — which is also why §11a finds Chen sitting on Qin's non-specific term. But the regression identifies affinity as the *predictor*; it does not prove that mechanism over residual container sorption, which would also scale with hydrophobicity.

Reproduce: `python3 scripts/chen_pfna_inversion.py`

Part IV — Using a binding constant

12. The bridge from K_d to a partition coefficient

A model wants capacity, not affinity. Here is the exact conversion.

Albumin at total molar concentration P, with n equivalent sites of dissociation constant K_d. Langmuir occupancy gives bound ligand per litre of solution:

```
C_bound = n · P · C_w / (K_d + C_w)
```

A partition coefficient is bound ligand per unit **sorbent** over free ligand per unit water. On a mass basis the sorbent is P·M_alb/1000 kg protein per litre, so:

```
D = [C_bound / (P·M_alb/1000)] / C_w  =  1000·n / [M_alb·(K_d + C_w)]
```

P cancels. The partition coefficient does not depend on how much albumin is in the tube — only on how many sites each albumin carries and how tightly they hold. In the trace limit (C_w << K_d):

> ***log D = log K_a + log n - 1.823*** *(mass basis, L_water/kg_protein)* ***log D = log K_a + log n - 1.688*** *(volume basis, L_water/L_protein)*

```
import math
M_ALB = 66463.0          # g/mol

def logD_from_ka(ka_M, n=1, basis="mass"):
    shift = math.log10(M_ALB / 1000.0)
    if basis == "volume":
        shift -= math.log10(1.0 / 0.733)      # albumin density 1.364 kg/L
    return math.log10(ka_M) + math.log10(n) - shift

ka = 1.0 / (2.57e-6)      # Chen 2025 PFOA
for n in (1, 3, 6):
    print(f"n={n}: log D = {logD_from_ka(ka, n):.2f} (mass) "
          f"{logD_from_ka(ka, n, 'volume'):.2f} (volume)")
```

Three groups write this same relation. Our derivation; Qin's mass-balance model, whose albumin term `C_max/(K_d + C_w) + D_ns` is this expression (its two C_max columns recover the pure-albumin molar concentration to 0.3%); and Chen 2025's Equation 4, published independently. It is not a modelling choice — it is the definition of a partition coefficient applied to a saturable site.

What the conversion needs that affinity papers rarely report. K_a and D are strictly proportional, log-log slope exactly 1. Everything a model needs is in the **intercept**, which is set by the molar-mass basis and by **n**. Moving n from 1 to 6 shifts log D by 0.78 with no new measurement.

Run it backwards and it becomes a diagnostic. Ask what stoichiometry would reconcile the tightest reported K_a with Fischer's directly measured log D: **n = 1.01 for PFOA, 1.12 for PFHpA.** Two completely different kinds of measurement agree at one binding site, with nothing fitted.

13. What happened when we used it

Short version: **albumin binding turned out not to explain what we hoped.**

Two published mass-balance models (Qin 2023, Armitage 2021) predict how much PFAS gets into cells in an in vitro assay. Both over-predict against 29 measured points. Albumin carries 19–94% of the sorptive capacity in those assays, so it looked like the obvious culprit.

We substituted Fischer's measured albumin partition coefficient — the only one measured at a physiological ligand:protein ratio — into both models. **Both got worse** (Qin 3.43× -> 4.74×, Armitage 2.33× -> 5.09×). Fischer's value is *lower* than the models assume, which shrinks the albumin sink and pushes more dose into the cell, and the models already over-predicted.

Inverting the balance for the albumin coefficient that *would* fix them gives values **above** anything measured. The data and the residual point in opposite directions. **Albumin is where the sensitivity is and not where the error is.**

The same thing happened with cell protein, from the other direction. Tracking it down (`MBM_Qin_vs_Armitage.md` §12–14) ended at the **cell volume**, which is uncertain to 3.1× and is the only parameter that rescales every compound by the same factor.

One result from that work is worth keeping regardless of the models: **Armitage's convention that 3.5% of cell protein behaves as octanol gives a protein affinity 457–7,820× below the measured value.** A coefficient calibrated on neutral organics does not describe an anionic surfactant on protein.

14. What is still open

1. **Mouse albumin has never been measured.** The largest single gap.
2. **n is almost never reported** with a K_d, and it moves log D by up to 0.78. Where it has been checked, n ~ 1 is what makes the datasets consistent (§11a, §12) — but that is three compounds, not a rule.
3. **Nobody has measured a PFAS–albumin constant at a genuinely physiological *nu*** (< 0.01) across a wide compound set. Chen's 0.06–2 is the closest.
4. **The sulfonates do not reconcile as cleanly as the carboxylates** (§11a), and nobody has explained why their specific-to-non-specific separation is so much larger.
5. **Where exactly inside nCF 9–11 is the carboxylate optimum?** §9a puts the plateau at 10–11 and rules out 6–8, but the top is flat and no study publishes per-compound error bars on that stretch.

6. **Where is the sulfonate optimum?** No dataset here carries a C12 sulfonate, so all that can be said is nCF >= 9.

Resolved since the first draft: the apparent Chen-versus-Fischer conflict, which is the specific/non-specific regime difference (§11a); Chen's PFNA "inversion", which is not significant and is one symptom of a general compression of his chain-length profile (§11b); and the location of the carboxylate chain-length optimum, which is nCF 10–11 and not the nCF 6–8 that three datasets report (§9a).

Check your understanding

Work these before looking at the answers. They are ordered by difficulty.

1. A study reports K_a = 1×10^5 M^{-1} for PFHxA–HSA. What is K_d in µM?

2. Using the 600 µM albumin figure and one binding site, what bound fraction does that K_a predict? Is it consistent with PFHxA being highly bound in plasma?

3. Two papers report PFOS–HSA constants: 2.2×10^4 M^{-1} and 7.6×10^6 M^{-1}. They are the same paper. What explains the 345× difference?

4. A DSF study reports K_d = 790 µM for PFOA. Why should this not go into a PBPK model, and what is it still good for?

5. Convert Chen 2025's PFOS (K_d = 0.94 µM) to log D on the mass basis at n = 1 and n = 6. Which would you use for a mass-balance model, and what would you need to know to decide?

6. Rat albumin binds PFOA ~2× more tightly than human albumin. The rat clears PFOA ~1000× faster. Reconcile these.

7. Why does substituting a *better* albumin measurement make a mass-balance model *worse*?

8. Chen's albumin affinities span 1.52 log units where Fischer's span 3.02 — Chen's curve is squashed toward its mean. Chen's maximum sits at nCF 7 and Fischer's at nCF 10. Does the squashing explain the three-carbon difference?

9. Ryu 2024 measures unbound fractions in five tissues across four species and finds them falling monotonically all the way to nCF 13, with no optimum. Does that refute an albumin optimum at nCF 10?

Answers

1. K_d = $1/10^5$ M = 10 µM. (`kd_from_ka(1e5)`)

2. f_bound = 600/(600+10) = 98.4%. Consistent with "highly bound" but it would not reach >99% at one site — you would need n ~ 2 or more.

3. Different **sites** on the same protein. 2.2×10^4 is the tryptophan site, which the authors call the weakest of three; 7.6×10^6 is Sudlow site II. Neither is wrong; quoting one without naming the site is.

4. DSF measures affinity at the protein's melting transition (~60–70 °C), not at 37 °C, and 790 µM predicts only 82% bound at n=6 where PFOA is observed >99% bound. It is still good for **ranking** congeners and comparing species by one consistent method.

5. log D = log(1/0.94e-6) + log n - 1.823 -> 4.20 at n=1, 4.98 at n=6. To choose you need the stoichiometry, which Chen fits as B_max per compound — and you need to know whether your model's albumin term is on a mass or volume basis (a further 0.135 log units).

6. Binding affinity is not the rate-limiting step. Clearance is dominated by renal **transporters** and tubular reabsorption, which differ enormously between species. Albumin binding modulates how much is available to be filtered, but a 2× difference cannot produce a 1000× one — and it points the wrong way.

7. Because the measured value is *lower* than the model assumed, which shrinks the albumin sink and sends more dose to the cell — and the model already over-predicted cell uptake. A parameter can be the most *sensitive* one in a model and still not be the one that is *wrong*.

8. No, and this is the trap. Squashing a curve toward its mean is a strictly increasing transform of the y-axis, and an argmax is invariant under any such transform: rescale Fischer by 0.82, or 0.5, or 0.25, and its peak stays at nCF 10. Two

curves that differ only in scale have the same optimum. So the three-carbon gap has to come from somewhere else — in this case a failure in Chen's long-chain tail, where K_D reverses direction between nCF 12 and 13 by more than its own standard errors allow (§9a).

9. No. Albumin has a cavity, and a cavity has a depth, so it can have an optimum. Lipid does not. Past nCF 10 the long-chain PFCAs do not stop binding — they stop binding *albumin* and go on partitioning into membrane, and Chen's own membrane–water coefficient rises smoothly to nCF 15 to prove it. Tissue f_unbound sums both channels and so sees the sum keep rising. The same logic blocks the reverse inference: "half-lives keep rising with chain length, so albumin affinity must too" does not follow either.

Reading list

Start here

- **Chen et al. 2025**, *Environ Sci Technol* 59:82–91, 10.1021/acs.est.4c06734 — 60 PFAS, equilibrium dialysis at *nu* = 0.06–2, plus membrane partitioning. The best single dataset. Its Eq. 4 is the K_d -> D bridge.
- **Qin et al. 2024**, *Environ Sci Technol* 58:9954–9966, 10.1021/acs.est.3c10824 — separates specific from non-specific binding, and settles the species question on whole plasma. Open access.

The site problem

- **Chen & Guo 2009**, *Arch Toxicol* 83:255–261, 10.1007/s00204-008-0359-x — the 5 × 3 compound-by-site grid, and the fatty-acid controls.

Species

- **Starnes et al. 2024**, *Toxicol Sci* 199:132–149, 10.1093/toxsci/kfae028 — four albumins, one method.
- **Ryu et al. 2024**, *Environ Sci Technol* 58:14641–14650, 10.1021/acs.est.4c04050 — whole tissue, four organisms, two mouse strains.

Structure and the physiological ratio

- **Fischer et al. 2024**, *Environ Sci Technol*, 10.1021/acs.est.3c07415 — log D at *nu* <= 0.004, 27 compounds, three protein targets.

Context on how much the conditions matter

- **Gao et al. 2025**, *Environ Sci Technol* 59:26716–26728, 10.1021/acs.est.5c08779 — the ionic strength series.
- **Starnes et al. 2025**, *Toxicol Sci* 203:67–78, 10.1093/toxsci/kfae124 — 65 PFAS by DSF; read as a ranking, not as affinities.

Where the data and code live

file	what
`db/protein_binding.csv`	865 rows, 66 study-years. **Group by study before averaging** — one study supplies 320 rows
`scripts/check_protein_binding.py`	validation; reproduces published statistics from three papers
`scripts/binding_kd_physiological_bound.py`	§4 View 2 and the §11 scoring table
`scripts/kd_to_partition.py`	§12, the full derivation and diagnostics
`scripts/binding_species_and_structure.py`	§9–10
`scripts/chen_fischer_reconciliation.py`	§11a
`scripts/chen_pfna_inversion.py`	§11b

file	what
`scripts/albumin_chain_length_optimum.py`	§9a, the optimum adjudication
`scripts/add_chen2025_tableS5.py`	the §9/§11 dataset
`report/REPORT.md` §6.1–6.3	the same material as evidence in the species argument
`report/MBM_Qin_vs_Armitage.md` §12–14	§13 in full